| SANTA CLARA UNIVERSITY | ECEN 123/MECH 143/CSEN 123 Winter 2026 | Andy Wolfe |
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# Final Project

## PROBLEM STATEMENT

After 6 hours of lab, I need a drink….

## PROBLEM DESCRIPTION

1. Your robot must autonomously complete a sequence of tasks. As of now, we do not have drawings of the final course, but you will be able to examine it and measure it for several weeks before the final demo.
2. There is a starting point near the right side of the course where the black line starts on the drawing. Your robot needs to start in that area.
    a. First you must navigate through the streets to the bar (which is in a back alley One possible path is partially marked with 1" wide black tape. Your robot must move through this corridor.
    b. Once you get around the first 2 turns, you get to the bar. To clock in at work, you need to push the green button. The light should turn off. You are now on the clock. You have 60 seconds to earn some $$$.
    c. You then need to check the order board next to the button. It will show you which beer has been ordered next. They are color coded. Red, Blue, Green, Yellow.
    d. To pour a beer, you must pull the corresponding tap handle forward 20°. To stop pouring, push it back to vertical. You only get one chance to pour each beer. You are a professional, you should get it right the first time.
    e. Like the real world, each tap may fill glasses at a different rate. Also, the fill rate for any given tap may change over time, but not during the 60 second run. Also, each beer is a slightly different color.
    f. If you underfill a beer, you don't get full points. If you spill the beer you don't get full points.
3. Teams will be given 2 runs on competition day. Best run counts.
4. If you need to reset the course electronics, you can power cycle it using the power strip switch on the back.

## GENERAL RULES

1. Do not do anything to alter or damage the course.
2. The competition will be held at the scheduled final exam time for the course: **March 19, 2026 at 09:10**. All team members are required to be in attendance and on time to the event.
3. Robots will be inspected prior to competing to ensure they meet the rules and specifications. We will arrange for a pre-inspection during office hours if you have anything you want evaluated in advance.
4. You can get 90% credit for any part of the lab up pouring your first beer if you demo each part to Dr. Wolfe or one of the TAs more than 12 hours prior to the final competition. For 100% credit you need to perform these tasks during the final competition.
5. Teams/robots that do not follow the rules and specifications will be disqualified. Breaking stuff that other groups need is particularly frowned upon. Report any damage or course malfunctions ASAP so they can be addressed.
6. Changes to competition format, rules, and/or outline will be posted to Camino.
7. Competition scoring and judgment will be final.
8. Questions about the competition and/or rules can be emailed to awolfe@scu.edu.
9. This document and the project rules are subject to minor changes at the discretion of the TA's and professor as needed.
10. If you think you may have a drinking problem, campus resources are available here.

## ROBOT DESIGN RULES

1. Robots must be autonomous. This means that the robot cannot receive any input from team members or devices not aboard the robot. This includes, and is not limited to, any forms of wireless transmission.
2. Only one robot is allowed per team.
3. It is the team's responsibility to design, build, and test the robot. You must write all code other that drivers supplied by component manufacturers or open source.
4. Your robot can be any size. It must fit through the field. You can cut the white platform I supply or substitute for it if you wish.
5. The power supply for the robot's motors must be the provided 9.6V battery, which must be carried on-board. You may use the 9V battery to power other components.
6. The robot should not damage the arena or anything in it.
7. Use of toxic, flammable, or otherwise harmful/dangerous substances is strictly prohibited. Non-toxic, non-combustible gases or liquids (i.e. water, compressed air, CO2) can only be used with prior approval from Dr. Wolfe. Approval will require:
    a. Documenting all components that will be under pressure
    b. Presenting a complete design for all hydraulics.
8. Use of Lego®, K'NEX®, or other off-the-shelf kits is not allowed.
9. Teams need not spend any money on their robots. Scrap parts are fine. If you do spend money, no expenses will be reimbursed.
10. Any and all of the components provided in the lab portion of ECEN 123/MECH 143/CSEN 123 may be incorporated into the design. Extra copies of these parts may be available. We also have a few larger motors and other parts. Check with the TAs/Prof. Please do not take any electronic parts without asking first. You may take fasteners as needed, but do not hoard them.
11. Judges reserve the right to terminate a robot's run, declare disqualification, or give instruction when necessary (e.g. if an unsafe event develops, or if the arena is in jeopardy).

12. The robot must start with the push of a single button. You can perform any calibration procedures you choose prior to starting.

**ARENA**

1. The arena floor is an unpainted wooden surface with black lines. It has 2.5" or higher walls. It will get dirty. There may be some small gaps in the boards. Lighting conditions will change throughout the day. Your robot needs to deal with it.
2. A 1" wide black masking tape line will mark portions of the route approximately as described above.
3. The finalized arena will be available to students for practice and testing purposes beginning about 3-4 weeks before the final. (likely in SCDI 4019). You can access the course at any time.
4. Robots must deal with lighting conditions present at the competition venue.

**There is no final lab report.**

Your team may keep your robot – but all unused parts and tools that are undamaged must be returned.

## COMPETITION RULES

1. The order of teams in competition will be decided at random, either through volunteers or conscription.
2. Robots will undergo inspection prior to competing
3. Each round is 3 minutes long, *maximum.*
4. Each team will get a turn to run their robot in Round 1.
5. As long as there is time (and we don't get bored), groups will be able to retry.
6. Only the best round for each team will be scored.
7. Teams *will* be allowed to repair, modify, tweak, or otherwise change their robot when not competing (i.e. after/in between runs).
8. If a robot fails to move for 20 seconds at any point of its run, it will be disqualified for that round. Useful movement of actuators other than drive wheels counts as moving. If the judges feel that a robot is doomed to fail on a round, they may end it.
9. Robots found to damage the arena will be disqualified for that round.
10. Teams will not be allowed to modify the arena in any way at any time.
11. Teams will be allowed to position their robot at the **START** prior to a run and press the start button. All other actions of the robot must be autonomous.
12. Bonus points for high scores.

## SCORING

Competition scoring will be based on the following criteria:

1. **Streets** – 15 points each for traversing the streets to the bar. 45 points total.
2. **Start Button** – 15 points pressing the start button and turning off its light.
3. **Taking the order** – 50 points for reading the order board the first time and going to the correct tap.
4. **Pouring the beer** – 20 points for starting to pour the correct beer. .
5. **Fulfilling orders** – You get points for properly filling the beer. Perfectly filled beer 25 points. One row short – 20 points, 2 rows short or spilled – 15 points. You can get points for up to 4 beers – but you need to check the order board each time to get the correct beer.
6. **Celebration** – 10 extra points if in the last 10 seconds of your round your robot does a celebration dance.
7. **Other Points** – Up to 20 additional points are awarded by the teaching staff for other project qualities. Teamwork, Project planning, Robot construction quality, Reliability and repeatability, creative design, Aesthetics.

*Penalties*: Major violations of the rules will result in disqualification of the run and a loss of points.

**DELIVERABLES AND SUGGESTED MILESTONES**

| Date | Suggested Milestone | Deliverable |
|---|---|---|
| 2/3 | Drivable, steerable platform | Lab 4 - Motors |
| 2/10 | Drive motors; wheels, "third wheel", battery; H-bridge, and Arduino permanently mounted. | |
| 2/15 | Plan for completing all tasks<br>Sketch of robot (with critical elements dimensioned) | |
| 2/22 | Circuit schematics including assigning Arduino pins for all sensors and actuators.<br>List of components and supplies to beg, borrow, make or buy. | Approval requests must be submitted for any high-power circuits or actuators and any gases or hydraulics. |
| 2/26 | Basic Line Following | Lab 8 – line following |
| 3/1 | Mechanical drawings for all additional components.<br>Plan and schedule for fabrication and/or assembly | |
| 3/6 | Assemble any additional actuators. Build drive circuits and basic motion code. | |
| 3/8 | Line following code completed and tested for actual field layout. | |
| 3/10 | Mount all sensors and actuators in final locations. | |
| 3/12 | Traverse entire course | |
| 3/14 | Reading orders | |
| 3/16 | Beer pouring | |
| 3/19 | | Final Demo |

**SAMPLE VIDEOS**

Below are some videos of similar competitions done by undergraduate mechatronics classes.

http://www.youtube.com/watch?v=hG9tZ9O60dA
http://www.youtube.com/watch?v=1hk-LV1laK8&feature=related
http://www.youtube.com/watch?v=SKRm2R38Flg

Some potential sources of materials and parts:

Digi-Key:
http://www.digikey.com/

Mouser Electronics:
http://www.mouser.com/

Halted Electronics:
http://www.excesssolutions.com/

McMaster Carr
http://www.mcmaster.com/

TAP Plastics:
1212 The Alameda
San Jose, CA 95126

Grainger Industrial Supply
1190 Kern Ave
Sunnyvale, CA 94085

Polulu Electronics
https://www.pololu.com/

Solarbotics
https://solarbotics.com/

Adafruit
http://www.adafruit.com/

Parallax
http://www.parallax.com/

Sparkfun
https://www.sparkfun.com/